Job Recruitment and Internship Platform

Section 5: Quality Attribute Scenarios

5.1 Introduction

This section gives six examples of how the Job Recruitment and Internship Platform should behave in different situations. Each example uses the same six parts: source, stimulus, artifact, environment, response, and response measure. They cover the main features (job applications, CV management, candidate matching, and interview scheduling), the users (applicants, recruiters, and administrators), and the services the platform depends on (email notifications and CV text extraction).

5.2 Overview

ID	Quality Attribute	Main Concern	Related Users / Services
QAS-1	Performance	Fast job search	Applicant
QAS-2	Availability	Email failure must not stop applications	Email Service
QAS-3	Security	Protect CVs and candidate data	Applicant, Recruiter
QAS-4	Modifiability	Replace the CV text-extraction tool easily	Apache Tika
QAS-5	Scalability	Handling a sudden rise in users	All users
QAS-6	Usability	Easy first-time application	Applicant

5.3 Scenarios

QAS-1: Performance (Job Search)

Part	Description
Source	Applicant
Stimulus	Searches for jobs using a keyword and filters (location, field)
Artifact	Job search function
Environment	Normal operation, 200 concurrent users, 50,000 active vacancies
Response	The platform returns matching vacancies, sorted by relevance
Response Measure	95% of search requests complete in under 500 ms, and 99% in under 1 second

QAS-2: Availability (Email Service Failure)

Part	Description
Source	Email Service
Stimulus	The service becomes unavailable while applicants submit applications
Artifact	Notification module
Environment	Normal operation, 100 applications submitted per hour
Response	The platform saves every application, places the emails in a queue, and retries sending automatically
Response Measure	0% of applications are lost, and 99% of emails are sent within 15 minutes after recovery.

QAS-3: Security (Unauthorized Access to CVs)

Part	Description
Source	Malicious user (logged-in applicant or unauthenticated user)

Part	Description
Stimulus	Tries to open another applicant's CV or a recruiter's candidate list by changing the URL or ID
Artifact	CV storage and candidate management modules
Environment	Normal operation, over the internet
Response	The platform denies access (HTTP 403), records the attempt in the audit log, and alerts the administrator after 5 failed attempts
Response Measure	100% of 500 test attacks are blocked and logged within 1 second.

QAS-4: Modifiability (Replacing Apache Tika)

Part	Description
Source	Developer
Stimulus	Wants to replace Apache Tika with another CV text-extraction tool
Artifact	CV processing component (Tika integration)
Environment	Design time, during a planned release
Response	The developer changes only the CV adapter. Matching, search, and notification modules stay unchanged
Response Measure	Change is completed in 3 person-days or less. Only 1 component is modified. 100% of regression tests pass after the change

QAS-5: Scalability (Job Fair Traffic Spike)

Part	Description
Source	Applicants and recruiters
Stimulus	Traffic grows from 500 to 5,000 concurrent users during a university job fair
Artifact	Web and application servers

Part	Description
Environment	Peak load, normal operation
Response	The platform adds server instances automatically and balances the load between them
Response Measure	Extra capacity is ready within 5 minutes. 95% of requests complete in under 1 second. Error rate stays below 1%

QAS-6: Usability (First-Time Applicant)

Part	Description
Source	First-time applicant
Stimulus	Wants to apply for an internship and upload a CV
Artifact	Application and CV upload screens
Environment	Normal operation, on a laptop or mobile browser
Response	The platform guides the user step by step and shows a confirmation message after submission
Response Measure	90% of 10 test users finish the application in under 5 minutes without help.